CELL DIVISION

Introduction

Cell division is a fundamental biological process through which a parent cell divides to form daughter cells. It is essential for growth, repair, reproduction, and maintenance of genetic continuity. In multicellular organisms, cell division increases the number of cells, whereas in unicellular organisms it acts as a mode of reproduction. The process ensures that genetic material is accurately transmitted from one generation of cells to the next.

Cell Cycle

The cell cycle is the sequence of events through which a cell passes to divide. It includes a preparatory phase and a division phase, both of which are tightly regulated to maintain genetic stability.

Interphase

Interphase is the longest phase of the cell cycle during which the cell prepares for division. It consists of three stages. In the G_1 phase, the cell grows and carries out normal metabolic activities. In the S phase, DNA replication occurs, resulting in the formation of two sister chromatids for each chromosome. Although the DNA content doubles, the chromosome number remains unchanged. In the G_2 phase, the cell prepares for mitosis by synthesizing proteins and checking for DNA errors.

M Phase

The M phase is the phase of actual division. It includes karyokinesis, which is the division of the nucleus, followed by cytokinesis, which divides the cytoplasm to form two daughter cells.

Mitosis

Mitosis is a type of cell division in which the chromosome number remains the same in the daughter cells as in the parent cell. It occurs in somatic cells and produces two genetically identical daughter cells. This process is essential for growth, repair, and maintenance of tissues.

Stages of Mitosis

Prophase

During prophase, chromatin condenses into visible chromosomes, each consisting of two sister chromatids joined at the centromere. The nuclear membrane begins to disappear, and spindle fibres start forming. Centrosomes move toward opposite poles of the cell.

Metaphase

In metaphase, chromosomes align at the equatorial plane of the cell. Spindle fibres attach to the centromere through kinetochores. This is the stage where chromosomes are most condensed and clearly visible, making it ideal for study.

Anaphase

During anaphase, the centromeres split, and sister chromatids separate from each other. These chromatids, now called daughter chromosomes, move toward opposite poles of the cell.

Telophase

In telophase, chromosomes reach the poles and begin to decondense into chromatin. The nuclear membrane reappears, and nucleolus becomes visible again, completing nuclear division.

Cytokinesis

Cytokinesis is the division of cytoplasm following nuclear division. In animal cells, it occurs through the formation of a cleavage furrow, while in plant cells, a cell plate is formed. This results in the formation of two separate daughter cells.

Meiosis

Meiosis is a specialized type of cell division that reduces the chromosome number by half and produces haploid cells. It occurs in germ cells and is essential for sexual reproduction. Meiosis introduces genetic variation and ensures that the chromosome number remains constant across generations.

Meiosis I

Meiosis I is known as reduction division because it reduces the chromosome number.

Prophase I

Prophase I is the longest and most complex stage. Homologous chromosomes pair in a process called synapsis, forming bivalents. Crossing over occurs between non-sister chromatids, resulting in genetic recombination and variation.

Metaphase I

Homologous chromosome pairs align at the equatorial plane, and spindle fibres attach to them.

Anaphase I

Homologous chromosomes separate and move toward opposite poles, reducing the chromosome number to half.

Telophase I

Two haploid daughter cells are formed, each containing half the number of chromosomes.

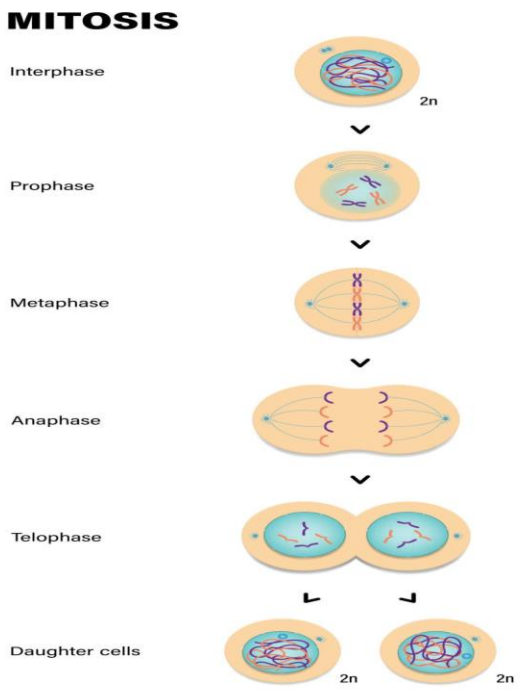
Meiosis II

Meiosis II is similar to mitosis and does not involve further reduction in chromosome number. Sister chromatids separate, resulting in the formation of four haploid daughter cells.

Significance of Cell Division

Cell division plays a crucial role in growth, repair, and reproduction. Mitosis ensures genetic stability by producing identical cells, while meiosis introduces variation and maintains chromosome number in sexually reproducing organisms. Together, these processes are essential for the continuity of life.

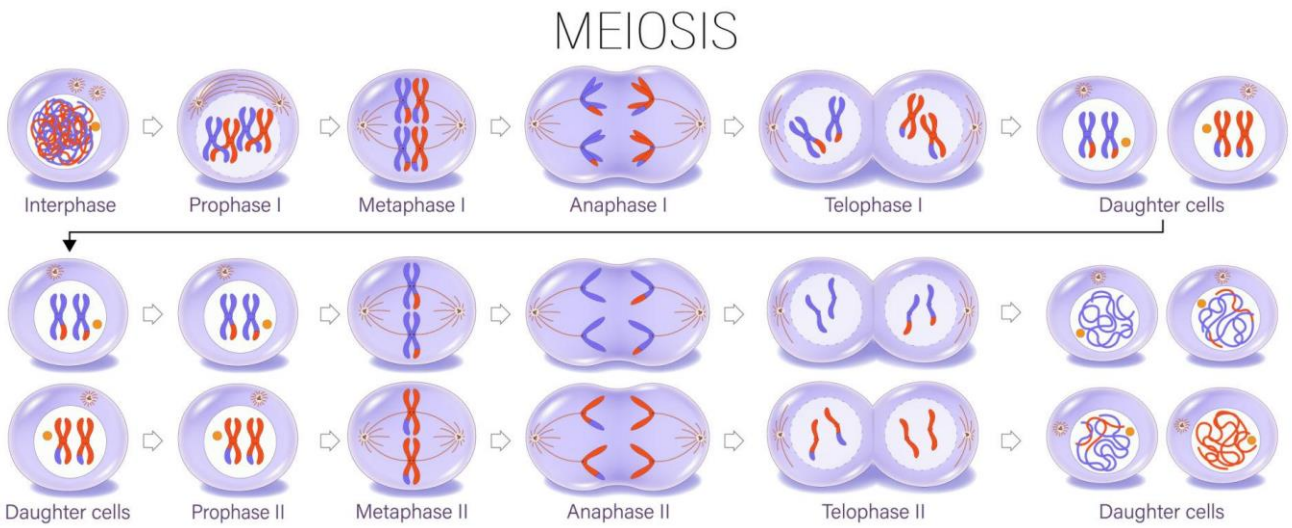
Mitosis Stages (PMAT)



Concept

This diagram shows the sequential stages of mitosis. Chromosomes condense in prophase, align in metaphase, separate in anaphase, and reform nuclei in telophase. Understanding this sequence is essential for NEET questions.

Meiosis Stages Diagram



Concept

This diagram illustrates meiosis I and II. The key feature is crossing over in prophase I and reduction of chromosome number in anaphase I. Meiosis results in four genetically different haploid cells.